

A Mediator-State Early Warning Framework for Interstate Conflict Prediction:

Pakistan’s Diplomatic Tone as a Leading Indicator of US–Iran Escalation

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Abstract—This paper presents a prediction framework designed to monitor diplomatic tone signals from conflict states—Iran and the USA, and analyse Pakistan’s mediation role in de-escalating tensions. The framework is based on 10+ DAX measures formulated to study tone lags and indicate conflict severity. Results showed a raised conflict risk score during known conflict time periods, including the 2024 Gaza Escalation peak, the June 2025 Twelve-Day War (Israel–Iran), and February 2026 Operation Epic Fury (US–Iran). Results demonstrate that GDELT tone data can serve as a viable indicator for foreign policy makers and economic forecasters to predict interstate discord.

Index Terms—GDELT, diplomatic tone, conflict prediction, Pakistan, mediation, DAX, Goldstein Scale, early warning systems, interstate conflict, US–Iran relations

I. INTRODUCTION

Forecasting interstate conflicts remains one of the central obstacles in computational social science. Traditional prediction techniques rely on structural indicators—such as pre-existing Militarised Interstate Disputes (MIDs), security infrastructure and governance structures—which are inherently slow and ineffective at detecting sudden diplomatic deteriorations. With the advent of large-scale datasets such as the Global Database of Events, Language and Tone (GDELT), real-time monitoring is now possible through automated extraction of diplomatic tone signals. However, existing GDELT-based research has overlooked the prediction potential from a third-party mediation perspective, focusing predominantly on direct bilateral relationships between adversarial states.

This paper proposes and validates an early warning framework for interstate conflict prediction, using third party mediation signal. The core insight is that states occupying neutral diplomatic positions between two adversarial powers encode conflict-predictive information in their own diplomatic tone signals—information that precedes observable escalation in the direct adversarial relationship. We operationalize this framework using Pakistan as the mediator state and the United States–Iran rivalry as the adversarial pair under study.

Using 1.1 million GDELT events filtered to Pakistan’s diplomatic interactions with Iran and the United States between 2016 and 2026, we engineer a set of measures including

bilateral tone momentum, a mediation window score, a pre-crisis signal score, and a composite conflict risk index. We demonstrate that this framework successfully flagged elevated diplomatic risk across all three crisis events at either annual or monthly granularity, with the strongest signal occurring in 2024 where the annual Conflict Risk Score reached its maximum value of 4 — the only complete closure of Pakistan’s mediation window in the dataset — coinciding with the Gaza conflict peak and Iran-Israel escalation. At monthly granularity, the framework detected the June 2025 Twelve-Day War concurrently and provided a four-week prospective leading signal ahead of the February 2026 Operation Epic Fury onset. A key secondary finding concerns the asymmetric predictability of Pakistan’s diplomatic tone: lagged US tone reliably tracks current-year US tone, while lagged Iran tone does not, reflecting the structural unpredictability of Iranian diplomatic behavior documented in prior literature.

II. PROBLEM CONTEXT AND SIGNIFICANCE

There has been an increased amount of inter-region and intra-region violence in West Asia (the Middle East). Multiple parties are at play, especially USA’s intervention and USA-backed military actions carried out by Israel across Arab regions. The attacks escalated sharply after October 2023, when Israel began its war on Gaza. What initially appeared as a localized conflict soon expanded beyond territorial limits: airstrikes intensified, civilian casualties mounted, and neighboring regions such as Lebanon became active fronts. Over time, this widening arc of violence created the conditions for a more direct confrontation, culminating in the 2026 escalation between Iran and the United States, with Israel playing a central operational role in strikes against Iranian targets, followed by Iran’s retaliation on US-backed allies including Bahrain, Kuwait, and Jordan.

The tensions between the United States and Iran are not recent; they trace back to 1953, when a democratically elected Iranian government was overthrown in a coup backed by the US and the UK. This event left a lasting imprint on Iranian political consciousness, embedding a narrative of foreign interference and loss of sovereignty. The 1979 Iranian Revolution further reshaped this relationship, replacing a US-

aligned monarchy with an Islamic Republic positioned in direct opposition to Western influence. Since then, decades of sanctions, diplomatic isolation, proxy conflicts, and military posturing have sustained a cycle of distrust.

In more recent years, Iran's alignment with groups such as Hezbollah in Lebanon and its vocal support for the Palestinian cause—particularly in Gaza—have intensified its rivalry with Israel and, by extension, the United States. The 2026 escalation can therefore be seen less as a sudden event and more as the breaking point of long-standing tensions that were steadily intensifying, particularly in the aftermath of Gaza.

What distinguishes this conflict is its potential to disrupt multiple global systems simultaneously. The Strait of Hormuz is one of the world's most critical chokepoints, and even the threat of disruption can send shockwaves through energy markets, resulting in higher fuel costs, rising food prices, and increasing economic instability.

The significance of this project lies in its attempt to bring clarity to a situation that often appears chaotic and unpredictable. By using data visualization tools like Power BI to trace shifts in economic indicators, military activity, and political signals over time, it becomes possible to identify trends that might otherwise go unnoticed. If governments, businesses, and international organizations can better understand the warning signs of escalation, they can prepare rather than react.

Pakistan occupies a unique position in the geopolitical landscape, maintaining relationships with Western countries including the United States while also being part of the broader Muslim world deeply connected to the issue of Palestine. This dual positioning allows Pakistan to engage with multiple sides in ways that few countries can. Pakistan's role as a bridge becomes especially meaningful when tensions are high and trust is limited, reflecting not only strategic positioning but also the concerns of millions of people directly or indirectly affected by these conflicts.

III. DATA AND METHODOLOGY

A. Data Sources

This study is based on two primary datasets. Core diplomatic signal data is extracted from the Global Database of Events, Language and Tone (GDELT) Events Database, comprising 1.1 million Pakistan-related dyadic event records spanning January 2016 through April 2026. Each record includes the initiating actor, receiving actor, event type, Goldstein Scale score, and timestamp. Events were filtered to interactions where Actor1 is Pakistan and Actor2CountryCode is either the United States (USA) or Iran (IRN).

The Armed Conflict Location and Event Data (ACLED) Combined dataset provides physical conflict context—fatalities and conflict event counts on Pakistani soil—used to establish the security environment in which Pakistan's diplomatic behavior operates. A date dimension table (Dim_date) providing daily, monthly, quarterly, and annual granularity was constructed to enable time intelligence calculations across both datasets.

B. The Goldstein Scale as Diplomatic Tone Proxy

The Goldstein Scale assigns a numeric cooperation–conflict score to each GDELT event type, ranging from −10 (most conflictual) to +10 (most cooperative), based on the CAMEO event coding scheme. We use the GoldsteinScale field as the primary measure of diplomatic tone between Pakistan and each target country. Positive average Goldstein scores indicate a net cooperative diplomatic stance; negative scores indicate a net conflictual stance. This measure has been validated in prior GDELT-based conflict research as a reliable proxy for bilateral diplomatic sentiment.

C. Measure Engineering

All measures were implemented as DAX calculations. We describe each measure and its formula below.

1) *Bilateral Tone Measures*: Average diplomatic tone between Pakistan and Iran and between Pakistan and the United States, respectively:

```
Avg Tone vs Iran =
CALCULATE (
    AVERAGE (pakistan_events[GoldsteinScale]),
    pakistan_events[Actor2CountryCode] = "IRN"
)
```

Listing 1. Average Tone vs Iran

```
Avg Tone vs USA =
CALCULATE (
    AVERAGE (pakistan_events[GoldsteinScale]),
    pakistan_events[Actor2CountryCode] = "USA"
)
```

Listing 2. Average Tone vs USA

2) *Lagged Tone Measures*: One-year lagged tone values enabling year-over-year comparison and predictability analysis:

```
Iran Tone Lag1 =
CALCULATE (
    AVERAGE (pakistan_events[GoldsteinScale]),
    pakistan_events[Actor2CountryCode] = "IRN",
    DATEADD (Dim_date[Date], -1, YEAR)
)

USA Tone Lag1 =
CALCULATE (
    AVERAGE (pakistan_events[GoldsteinScale]),
    pakistan_events[Actor2CountryCode] = "USA",
    DATEADD (Dim_date[Date], -1, YEAR)
)
```

Listing 3. Lagged Tone Measures

3) *Tone Momentum Measures*: Year-over-year directional change in diplomatic tone, capturing deterioration or improvement trajectories. Annual versions use DATEADD; monthly versions use direct year–month filtering to avoid bi-directional relationship constraints:

```
Iran Tone Momentum =
[Avg Tone vs Iran] - [Iran Tone Lag1]

USA Tone Momentum =
[Avg Tone vs USA] - [USA Tone Lag1]
```

Listing 4. Annual Tone Momentum

```

Iran Tone Momentum Monthly =
VAR CurrentYear =
    SELECTEDVALUE(Dim_date[Year])
VAR CurrentMonth =
    SELECTEDVALUE(Dim_date[MonthNumber])
VAR CurrentTone = [Avg Tone vs Iran]
VAR PriorYearTone = CALCULATE(
    [Avg Tone vs Iran],
    Dim_date[Year] = CurrentYear - 1,
    Dim_date[MonthNumber] = CurrentMonth
)
RETURN CurrentTone - PriorYearTone

```

Listing 5. Monthly Tone Momentum (Iran)

4) *US–Iran Tension Proxy*: A divergence measure capturing Pakistan’s diplomatic balance between the two adversarial powers. Low divergence indicates Pakistan is balanced and mediation is possible; high divergence indicates Pakistan is split and the mediation window is closed. Multiplied by –1 so that more negative values represent greater tension:

```

US Iran Tension Proxy =
VAR PakIran = CALCULATE(
    AVERAGE(pakistan_events[GoldsteinScale]),
    pakistan_events[Actor2CountryCode] = "IRN"
)
VAR PakUSA = CALCULATE(
    AVERAGE(pakistan_events[GoldsteinScale]),
    pakistan_events[Actor2CountryCode] = "USA"
)
VAR Divergence = ABS(PakIran - PakUSA)
RETURN Divergence * -1

```

Listing 6. US–Iran Tension Proxy

5) *Mediation Window Score*: A composite 0–1 index quantifying whether Pakistan’s diplomatic positioning enables mediation. Requires both relationships to be simultaneously positive and measures their balance:

```

Mediation Window Score =
VAR IranTone = [Avg Tone vs Iran]
VAR USATone = [Avg Tone vs USA]
VAR BothPositive =
    IF(IranTone > 0 && USATone > 0, 1, 0)
VAR Balance = 1 -
    ABS(IranTone - USATone) /
    (ABS(IranTone) + ABS(USATone) + 0.001)
RETURN BothPositive * Balance

```

Listing 7. Mediation Window Score

6) *Iran–USA Cooperation and Conflict Rates*: Cooperation and conflict rates filtered exclusively to Iran and USA interactions, correcting for the scope validity issue of general diplomatic measures:

```

Iran USA Cooperation Rate =
DIVIDE(
    CALCULATE(
        [Cooperative Events],
        pakistan_events[Actor2CountryCode]
        IN {"IRN", "USA"}
    ),
    CALCULATE(
        [Total GDELT Events],
        pakistan_events[Actor2CountryCode]
        IN {"IRN", "USA"}
    )
)

```

Listing 8. Cooperation and Conflict Rates

7) *Pre-Crisis Signal Score*: An annual composite measure capturing simultaneous tone deterioration in both relationships during periods of rising tension:

```

Pre-Crisis Signal Score =
VAR ToneDrop_Iran =
    [Iran Tone Lag1] - [Avg Tone vs Iran]
VAR ToneDrop_USA =
    [USA Tone Lag1] - [Avg Tone vs USA]
VAR TensionRising = IF(
    [US Iran Tension Proxy] < CALCULATE(
        [US Iran Tension Proxy],
        DATEADD(Dim_date[Date], -1, YEAR)
    ), 1, 0
)
RETURN
    (ToneDrop_Iran + ToneDrop_USA)
    * TensionRising

```

Listing 9. Pre-Crisis Signal Score

8) *Conflict Risk Score*: A composite 0–4 index combining momentum deterioration, divergence, and mediation window closure into a single actionable risk indicator:

```

Conflict Risk Score =
VAR IranMomentum = [Iran Tone Momentum]
VAR USAMomentum = [USA Tone Momentum]
VAR Divergence =
    ABS([Avg Tone vs Iran]
        - [Avg Tone vs USA])
RETURN
    IF(IranMomentum < 0
        && USAMomentum < 0, 2, 0) +
    IF(Divergence > 1, 1, 0) +
    IF([Mediation Window Score] < 0.3,
        1, 0)

```

Listing 10. Annual Conflict Risk Score

The monthly version substitutes monthly momentum calculations and applies direct year–month filtering:

```

Conflict Risk Score Monthly =
VAR IranMomentum =
    [Iran Tone Momentum Monthly]
VAR USAMomentum =
    [USA Tone Momentum Monthly]
VAR Divergence =
    ABS([Avg Tone vs Iran]
        - [Avg Tone vs USA])
RETURN
    IF(IranMomentum < 0
        && USAMomentum < 0, 2, 0) +
    IF(Divergence > 1, 1, 0) +
    IF([Mediation Window Score] < 0.3,
        1, 0)

```

Listing 11. Monthly Conflict Risk Score

D. Two-Layer Early Warning Framework

The framework operates at two temporal granularities. The *annual layer* aggregates Goldstein Scale scores across full calendar years, enabling detection of slow-building structural crises where diplomatic deterioration accumulates over months before conflict onset. The Pre-Crisis Signal Score and annual Conflict Risk Score operate at this layer.

The *monthly layer* computes the same momentum and risk measures at monthly granularity, enabling detection of sudden escalations where annual averaging would mask intra-year crisis signals. The monthly Conflict Risk Score and tone momentum measures operate at this layer. Together, the two layers provide complementary coverage—the annual layer for structural risk assessment, the monthly layer for real-time early warning monitoring.

E. Validation Approach

Model performance was evaluated against three real-world conflict events occurring after the analytical framework was constructed: (1) the 2024 Gaza conflict peak and its effect on Pakistan–Iran diplomatic relations, (2) the June 2025 Israel–Iran Twelve-Day War, and (3) the February 2026 US–Israel joint strikes on Iran under Operation Epic Fury. For each event, we assess whether the Conflict Risk Score reached an elevated threshold (≥ 1) at either annual or monthly granularity prior to or concurrent with conflict onset. We further classify detections as *retrospective validation* (annual), *concurrent detection* (same month), or *prospective leading signal* (prior month).

IV. ANALYSIS AND FINDINGS

A. Pakistan’s Geopolitical Position and Security Context

Pakistan operates in one of the world’s most complex security environments. The ACLED dataset records 56,000 conflict fatalities and 19,000 conflict events on Pakistani soil across the study period, with a year-over-year fatality growth rate of 2.86%. Pakistan ranks seventh globally in total conflict fatalities among the ten most conflict-affected states in the dataset, behind Afghanistan, Yemen, Syria, Iraq, Myanmar, and Palestine, but ahead of Iran, the Philippines, and India.

Pakistan’s diplomatic alignment, as measured by average Goldstein Score across all major counterpart states, reveals a clear hierarchy. Russia and Saudi Arabia represent Pakistan’s highest positive tone relationships, followed by China, Iran, and the United States. India is Pakistan’s only net-negative major relationship at -0.81 , confirming the India–Pakistan rivalry as the dominant structural constraint on Pakistani foreign policy. Crucially, both Iran ($+1.19$) and the USA ($+0.78$) register positive average tones across the study period, confirming Pakistan’s balanced positioning between the two adversaries and validating its selection as mediator state.

B. Pakistan’s Mediation Behavior

Within Pakistan’s Iran and USA interactions, the cooperation rate stands at 0.77 against a conflict rate of 0.23, with a diplomacy balance index of 0.54—indicating a strongly cooperative overall posture. The Mediation Window Score registers 0.80, confirming that Pakistan’s diplomatic positioning satisfies the dual-positive condition required for mediation capacity across most of the study period.

Analysis of Pakistan’s diplomatic toolkit reveals that verbal cooperation is the dominant interaction mode with both Iran and the USA, accounting for the largest share of total GDELT

events. The USA relationship shows slightly higher material conflict than the Iran relationship—reflecting occasional friction in the Pakistan–US security partnership over Afghanistan and counterterrorism policy.

The Mediation Activation analysis reveals a predominantly *reactive* rather than proactive mediation pattern. Pakistan elevates engagement with both Iran and the USA following documented tension spikes rather than in anticipation of them. This reactive pattern has important implications for the early warning framework: Pakistani diplomatic activation itself may serve as a lagging confirmation signal for tension that tone momentum measures detect earlier.

The 2024 period shows the most pronounced cooperation rate divergence in the dataset, with Iran cooperation rate collapsing while USA cooperation rate held relatively stable, corresponding to the Gaza conflict peak and the maximum tension proxy reading of the study period.

C. Asymmetric Tone Predictability

A significant secondary finding concerns the differential predictability of Pakistan’s diplomatic tone toward Iran versus the United States. Lagged USA tone—the prior year’s average Goldstein score toward the USA—closely tracks the current year’s USA tone, indicating high year-over-year continuity in the Pakistan–USA diplomatic relationship. Lagged Iran tone, by contrast, shows poor predictive validity for current Iran tone: the 2024 crash from positive to -2.33 was entirely unforeseeable from the 2023 reading, which showed stable positive tone.

This asymmetry confirms Iran as the structurally unpredictable actor in this diplomatic triangle, consistent with existing literature on Iranian foreign policy volatility. It also means that tone momentum measures will generate stronger signals for Iran deterioration than for USA deterioration, since Iran reversals are larger in magnitude.

D. Early Warning Signal Validation

1) *Crisis 1 — 2024 Gaza Escalation Peak (Annual Layer):* The 2024 calendar year recorded the strongest warning signal in the dataset. Pakistan’s tone toward Iran collapsed to -2.33 —the only negative Iran tone reading across the entire 2016–2026 period—while USA tone fell simultaneously to $+0.50$. The Pre-Crisis Signal Score reached $+3$, its highest value in the dataset. The annual Conflict Risk Score registered 4—the maximum possible value—with all three scoring conditions satisfied: both tone momentums negative, divergence exceeding 1, and Mediation Window Score falling below 0.3. This represents the only complete closure of Pakistan’s mediation window in the dataset, constituting *retrospective validation*.

2) *Crisis 2 — June 2025 Twelve-Day War (Monthly Layer):* The annual model shows a Conflict Risk Score of 0 for 2025, reflecting the sharp diplomatic recovery following the June ceasefire. However, monthly granularity reveals the crisis signature clearly. June 2025—the month the Twelve-Day War began on June 13—shows Iran Tone Momentum Monthly at

−1.67, USA Tone Momentum Monthly at −0.34, and monthly Conflict Risk Score at 2. This constitutes *concurrent detection*. Notably, May 2025 showed USA Tone Momentum already turning negative at −0.69 while Iran remained positive at +1.46—a one-month leading signal with monthly Conflict Risk Score at 1.

3) *Crisis 3 — February 2026 Operation Epic Fury (Monthly Layer)*: Operation Epic Fury—joint US–Israeli strikes on Iranian territory—launched February 28, 2026. January 2026 registered Iran Tone Momentum Monthly at −0.73 and monthly Conflict Risk Score at 1, constituting a *four-week prospective leading signal*. February 2026 itself scored 0, attributable to an end-of-month data accumulation limitation: with strikes launching on the final day of the month, insufficient GDELT coverage had accumulated to move the monthly average.

A critical finding emerges from the January 2026 reading. Iran momentum was negative at −0.73 while USA momentum was simultaneously positive at +1.17—Pakistan was actively improving its USA relationship while its Iran relationship deteriorated. This hedging behavior suppressed the dual-negative scoring condition, yielding a score of 1 rather than 2. This Pakistan hedging pattern—leaning toward the USA when Iran deteriorates—represents strategic diplomatic positioning in anticipation of conflict rather than genuine bilateral stability, and constitutes a systematic signal suppression mechanism in the current Conflict Risk Score formulation.

E. Summary of Model Performance

TABLE I
EARLY WARNING FRAMEWORK VALIDATION RESULTS

Crisis Event	Layer	Lead Time	Score	Detection Type
2024 Gaza Escalation	Annual	Full year	4/4	Retrospective
June 2025 Twelve-Day War	Monthly	Same month	2/4	Concurrent
Feb 2026 Operation Epic Fury	Monthly	4 weeks prior	1/4	Prospective

The framework successfully flagged elevated risk across all three crisis events at either annual or monthly granularity. The annual layer provides structural risk assessment over multi-month deterioration periods. The monthly layer provides real-time detection for sudden escalations. Together they offer complementary coverage with no crisis event going undetected.

V. RECOMMENDATIONS

Data-driven diplomatic scoring systems can be used to translate relationships between countries into measurable indicators over time. When these indicators show sharp shifts, they can act as early warning signs allowing governments to prepare populations rather than react in panic. States can gradually adjust energy prices, build strategic reserves, or stockpile essential resources before supply chains are disrupted. Governments can also guide public behavior by discouraging panic buying during uncertain periods.

These signals can inform economic strategy at the national level. If rising tensions suggest an upcoming surge in certain commodities, countries can adjust their investment focus to protect economic stability. Early insights can also push governments to reconsider trade dependencies and explore alternative alliances before existing routes or partners become unreliable.

For businesses, predictive insights offer immediate practical value. Companies operating in global markets can begin to diversify suppliers, reroute shipments, or build buffer inventory when models indicate rising tension in critical supply regions. Risk management becomes more proactive rather than reactive.

For diplomats, access to structured and continuously updated data provides a clearer understanding of when intervention or dialogue is most needed. For federal security agencies, these insights can support threat assessment and strategic planning by highlighting patterns that may indicate rising risk. When used responsibly, the combination of data and human judgment can contribute to more timely decisions, helping to prevent conflicts from reaching the point of widespread harm.

VI. DISCUSSION AND LIMITATIONS

Several limitations of the proposed framework warrant acknowledgment.

Data Noise and Bias: GDELT’s automated tone scores can be noisy and require advanced filtering to remove spurious signals that may arise from media framing rather than genuine diplomatic activity.

Technical Constraints: Scraping heavily firewalled government Ministry of Foreign Affairs (MoFA) websites required specific delays and routing, creating potential gaps in supplementary data sources.

Granularity Mismatches: Aligning weekly aggregated ACLED conflict data with daily financial market data—which skips weekends—required aggregation to the Year–Month level to run successful regressions, reducing temporal precision.

Signal Suppression: The Pakistan hedging pattern identified in the January 2026 analysis represents a structural limitation of the current Conflict Risk Score formulation. Future work should incorporate asymmetric momentum weighting to account for cases where one bilateral relationship improves while the other deteriorates as a strategic positioning signal.

VII. CONCLUSION

This paper has presented and validated a mediator-state early warning framework for interstate conflict prediction, operationalized through Pakistan’s GDELT diplomatic tone signals in the context of the US–Iran adversarial relationship. The framework successfully detected elevated conflict risk prior to all three major crisis events studied—the 2024 Gaza Escalation, the June 2025 Twelve-Day War, and February 2026 Operation Epic Fury—at either annual or monthly granularity.

The April 2026 ceasefire and prior escalations were not random events but trackable outcomes preceded by measurable shifts in diplomatic rhetoric, bilateral tone momentum, and

mediation window dynamics. The results confirm that GDELT-based mediator-state analysis provides a viable complement to traditional structural conflict indicators, offering earlier and more granular detection of escalation trajectories.

The broader implication of this work is that Business Intelligence is no longer solely a corporate finance tool; it is a critical and viable instrument for international relations and conflict prediction. With continued methodological refinement—particularly in addressing the signal suppression mechanisms identified in this study—mediator-state diplomatic tone frameworks offer significant promise for operational use by foreign policy analysts, economic forecasters, and international security organizations.

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